Statistical Distributions

1: General Probability Distributions

"Given that $P\left(X = x\right) = \frac{k}{x}$, find the value of k."

2: Binomial Distribution

"I toss an unfair coin, with probability heads of 0.6, 10 times. What's the probability I see 5 heads?"

3: Cumulative Binomial Probabilities

"I toss an unfair coin, with probability heads of 0.6, 10 times. What's the probability I see at most 3 heads?"

Probability distributions 3

You are already familiar with the concept of **variable** in statistics: a collection of values (e.g. favourite colour of students in the room):

$\mathbf{x}$	red	green	blue	orange
$\mathbf{P(X = x)}$	0.3	0.4	0.1	0.2

If each is assigned a probability of occurring, it becomes a **random variable**.

✎ A **random variable $\mathbf{X}$ represents a single experiment/trial.** It consists of outcomes with a probability for each.

i.e. X is a random variable (capital letter), but x is a particular outcome.

$$P(X = x)$$

"The probability that…" …the outcome of the random variable X… …was the specific outcome x"

A shorthand for $P(X = x)$ is ✎ $p(x)$ (note the lowercase p).
It's like saying "the probability that the outcome of my coin throw was heads" ($P(X = heads)$) vs "the probability of heads" ($p(heads)$). In the latter the coin throw was implicit, so we can skip the '$X =$'.

Probability Distributions vs Probability Functions 4

There are two ways to write the mapping from outcomes to probabilities:

The "{" means we have a 'piecewise function'. This just simply means we choose the function from a list depending on the input.

As a function:

$$p(x) = \begin{cases} 0.1x, & x = 1, 2, 3, 4, \\ 0, & \text{otherwise.} \end{cases}$$

e.g. if $x = 2$, then the probability is $0.1 \times 2 = 0.2$

As a table:

$\mathbf{x}$	1	2	3	4
$\mathbf{p(x)}$				

The table form that you know and love.

Advantages of functional form:

Advantages of table form:

Example 5

The random variable X represents the **number of heads when three coins are tossed**.

Underlying Sample Space

Distribution as a Table

Distribution as a Function

Example Exam Question 6

(Hint: Use your knowledge that $\Sigma p\,(\ldots) = 1$)

Edexcel S1 May 2012

1. A discrete random variable X has probability function

$$P(X = x) = \begin{cases} k(1-x)^2, & x = -1, 0, 1, 2 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Show that $k = 1/6$. (3 marks)

x	-1	0	1	2
p(x)				

a

x	2	3	4	5
p(x)	0.1	0.3	0.2	0.4

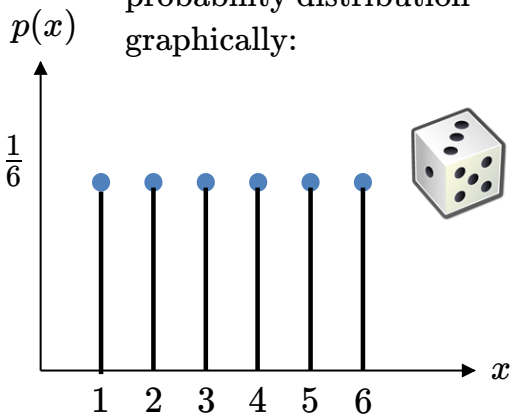
Determine:

$P(X > 3) =$

$P(2 \leq X < 4) =$

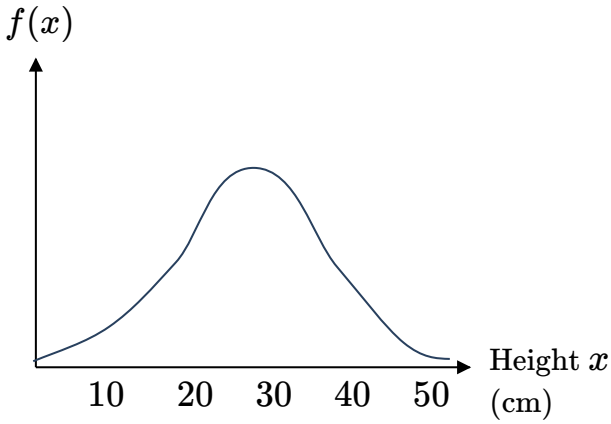
$P(2X + 1 \geq 6) =$

We can also represent a probability distribution graphically:



The throw of a die is an example of a **discrete uniform distribution** because the probability of each outcome is the same.

$p(x)$ for discrete random variables is known as a **probability mass function**, because the probability of each outcome represents an actual 'amount' (i.e. mass) of probability.



We can also have probability distributions for **continuous** variables, e.g. height.

However, the probability that something has a height of say **exactly** 30cm, is infinitely small (effectively 0).

$p(x)$ (written $f(x)$) for continuous random variables is known as a **probability density function**. $p(30)$ wouldn't give us the probability of being 30cm tall, but the amount of probability **per unit height**, i.e. the density. This is similar to histograms where frequency density is the "frequency per unit value". Just as an area in a histogram would then give a frequency, and area under a probability density graph would give a probability (mass).

You will encounter the **Normal Distribution** in Year 2, which is an example of a continuous probability distribution.

5. $P(X = x) = kx$ for $x = 1, 2, 3, 4$. Show that $k = 1/10$.
6. $P(X = x) = kx$ for $x = 1, 3$ and $k(x - 1)$ for $x = 2, 4$, where k is a constant.

(a) Find k.

(b) Find $P(X > 1)$.

5

6a

6b

7. The discrete random variable X has probability function

$$P(X = x) = \begin{cases} 0.1, & x = -2, -1 \\ \beta, & x = 0, 1 \\ 0.2, & x = 2. \end{cases}$$

(a) Find β.

(b) Construct a table giving the distribution of X.

(c) Find $P(-1 \leq X < 2)$.

a

b

c

10. Use the given probability distribution to find:
(a) $P(1 < X \leq 3)$ (1 mark);
(b) $P(X < 2)$ (1 mark);
(c) $P(X > 3)$ (1 mark).
11. A biased coin with $P(H) = 2/3$ is tossed until a head appears or four tosses have been made.
(a) Write down the distribution of the number of tosses S, in table form. (4 marks)
(b) Find $P(S > 2)$. (1 mark)

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

10a

10b

10c

11a

11b

13. Marie claims $P(X = x) = 2/x^2$ for $x = 2, 3, 4$ defines a probability distribution.
(a) Explain why it does not. (2 marks)
(b) Given the correct probability mass function is $P(X = x) = k/x^2$ for $x = 2, 3, 4$, find the exact value of k. (2 marks)

a

b

Extension 13

[Challenge p88] The Independent Random Variables X and Y have probability distributions

$P(X = x) = \frac{1}{8}$, $x = 1, 2, 3, 4, 5, 6, 7, 8$ $P(Y = y) = \frac{1}{y}$, $y = 2, 3, 6$

Find $P(X > y)$.

Solution

Leftie Example 14

Let's simplify the problem by using just 3 people:

The probability a randomly chosen person is left-handed is 0.1. If there is a group of 3 people, what is the probability that:

a) All 3 are left-handed.

b) 0 are left-handed.

c) 1 person is left-handed.

d) 2 people are left-handed.

d

a

b

c

Let's try to generalise!

If there were x 'lefties' out of 3, then we can see, using the examples, that the probability of a single matching outcome is $0.1^x \times 0.9^{3-x}$.

How many rows did we have each time? In a sequence of three L's and R's, there are "3 choose x", i.e. $\binom{3}{x}$ ways of choosing x of the 3 letters to be L's. Therefore the probability of x out of 3 people being left handed is:

$$\binom{3}{x} 0.1^x 0.9^{3-x}$$

You can model a random variable X with a binomial distribution $B(n, p)$ if
- there are a fixed number of trials, n,
- there is a fixed probability of success, p
- the trials are independent of each other

If $X \sim B(n, p)$ then:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

← In our example, 'success' was 'leftie'.

← r is the number of successes out of n.

← "$\sim$" means "has the distribution"

On a table of 8 people, 6 people are left handed.

a) Suggest a suitable model for a random variable X: the number of left-handed people in a group of 8, where the probability of being left-handed is 0.1.

b) Find the probability 6 people are left handed.

c) Suggest why the chosen model may not have been appropriate.

a

b

c

In general, choosing a well-known model, such as a Binomial distribution, makes certain **simplifying assumptions**. Such assumptions simplifies the maths involved, but potentially at the expense of not adequately modelling the situation.

Further Examples 16

The random variable $X \sim B\left(12, \frac{1}{6}\right)$. Find:

a) $P(X = 2)$

b) $P(X = 9)$

c) $P(X \leq 1)$

a

b

c

Mental Tip: The two powers add up to n.

Tip: Remember the two 'edge cases':
$P(X = 0) = (1-p)^n$
$P(X = n) = p^n$

Edexcel S2 June 2010 Q6

A company claims that a quarter of the bolts sent to them are faulty. To test this claim the number of faulty bolts in a random sample of 50 is recorded.

(a) Give two reasons why a binomial distribution may be a suitable model for the number of faulty bolts in the sample. (2)

a

Test Your Understanding 17

1 $X \sim B(6, 0.2)$

What is $P(X = 2)$?

1a

What is $P(X \geq 5)$?

1b

2 I have a bag of 2 red and 8 white balls. X represents the number of red balls I chose after 5 selections (with replacement).

a How is X distributed?

2a

b Determine the probability that I chose 3 red balls.

2b

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6. The probability that a switch is faulty is 0.08. A random sample of 10 switches is taken from the production line.

(a) Define a suitable distribution for the number of faulty switches and justify it. (2 marks)

(b) Find the probability that four switches are faulty. (2 marks)

a

b

7. A genetic marker is present in 4% of the population.

(a) State the assumptions needed to model the number of people with the marker in a sample of size n by a binomial distribution. (2 marks)

(b) Find the probability that exactly 6 in a sample of 50 have it. (2 marks)

a

b

8. A die lands on six with probability 0.3. Hannah rolls it 15 times.

(a) State the assumptions needed to model the number of sixes as binomial. Give the distribution. (2 marks)

(b) Find the probability of exactly four sixes. (2 marks)

(c) Find the probability of two or fewer sixes. (3 marks)

a

b

c

Often we wish to find the probability of a range of values.
For a Binomial distribution, this was relatively easy if the range was narrow, e.g. $P(X \leq 1) = P(X = 0) + P(X = 1)$, but would be much more computationally expensive if we wanted say $P(X \leq 6)$.

$$\text{If } X \sim B(10, 0.3), \text{ find } P(X \leq 6).$$

How to calculate

On Classwiz: Press Menu then 'Distributions'.
- Choose "Binomial CD" (the C stands for 'Cumulative'). Choose 'Variable'.
- $x = 6$, $N = 10$, $p = 0.3$

On CG50: Menu Statistics, Dist, Binomial, Bcd
- Set Data=variable, Lower=0, upper =6,
- Numtrial=10, p=0.3

On fx-9860GIII: Statistics, Dist, Binomial, Bcd
- Set Data=variable
- - $x = 6$, $N = 10$, $p = 0.3$

In all calculator types press "=" to see the results

Using tables (e.g. Page 204 of textbook)

Look up $n = 10$ and the column $p = 0.3$.
Then look up the row $x = 6$.
The value should be 0.9894.

Important Note: The tables only have limited values of p. You may have to use your calculator. You will need to use your calculator in the exam anyway.

The random variable $X \sim B(20, 0.4)$. Find:

Look up $n = 20, p = 0.4, x = 7$

$P(X \leq 7) =$
$P(X < 6) =$
$P(X \geq 15) =$

Note that the table requires $\leq$

To get this right, just say in your head "*What's the opposite of 'at least 15'?*". Hopefully you can see it's '*at most 14*'.

Given that $X \sim B(25, 0.25)$

$P(X = 6) =$
$P(X > 20) =$
$P(6 < X \leq 10) =$

X can be 7 to 10. So we want up to 10, with everything up to 6 excluded.

Quickfire Questions

Write the following in terms of cumulative probabilities, e.g. $P(X < 7) = P(X \leq 6)$

$P(X < 5) =$
$P(X \geq 7) =$
$P(X > 7) =$
$P(10 \leq X < 20) =$

$P(10 \leq X \leq 20) =$
$P(X = 100) =$
$P(20 < X < 30) =$
at least 30 =
greater than 30 =

More Challenging Example 23

Q

An awkward boyscout boy ventures into a girl guide camp. He asks 20 girls out on the date. The probability that each girl says yes is 0.3.
Determine the probability that he will end up with:
a) Less than 6 girls on his next date.
b) At least 9 girls on his next date.

The boy considers the evening a success if he dated at least 9 girls that evening.
He repeats this process across 5 evenings.
c) Calculate the probability that he had at least 4 successful evenings.

(Note: You won't be able to use your table for (c) as p is not a nice round number – calculate prob directly)

a

a

b

b

c

c

This is an interesting problem because the probability from a Binomial distribution is then used as the p of a **second** separate Binomial distribution.

Dealing with Probability Ranges 24

Q

A spinner is designed so that probability it lands on red is 0.3. Jane has 12 spins.
a) **Find the probability that Jane obtains at least 5 reds.**
Jane decides to use this spinner for a class competition. She wants the probability of winning a prize to be < 0.05. Each member of the class will have 12 spins and the number of reds will be recorded.
b) Find how many reds are needed to win the prize.

STEP 1: Represent the sentence using probability.

STEP 2: Ensure LHS involves $\leq$ inside probability.

STEP 3: Rearrange.

STEP 4: Use table backwards to find value corresponding to closest probability.

Test Your Understanding 25

Q At Camford University, students have 20 exams at the end of the year. All students pass each individual exam with probability 0.45. Students are only allowed to continue into the next year if they pass some minimum of exams out of the 20.
What do the university administrators set this minimum number such that the probability of continuing to next year is at least 90%?

Exercise 6C – Statistics Year 1 Pages 93-94 26

7. $X \sim B(50, 0.40)$. Find:
(a) the largest integer k for which $P(X \leq k) < 0.05$ (1 mark);
(b) the smallest integer r for which $P(X > r) < 0.01$ (2 marks).

a

b

8. $X \sim B(40, 0.10)$. Find:
(a) the largest integer k for which $P(X < k) < 0.02$ (1 mark);
(b) the smallest integer r for which $P(X > r) < 0.01$ (2 marks);
(c) $P(k \leq X \leq r)$ (2 marks).

a

b

c

9. In a town, 30% of residents listen to local radio. Ten residents are chosen at random. X counts those who listen.
(a) Suggest a suitable distribution for X and give your assumptions. (2 marks)
(b) Find the probability that at least half listen. (2 marks)
(c) Find the smallest s such that $P(X \geq s) < 0.01$. (2 marks)

a

b

c

10. A motor component factory has a defect rate of 5%. An officer inspects random samples of 50. Find the probability that the next sample contains:

(a) fewer than 2 defectives (1 mark);

(b) more than 5 defectives (2 marks).

(c) Production stops when defectives exceed d. Given production stops less than 5% of the time, find the smallest d. (2 marks)

a

b

c